

Literature-Already-Answered Packet
Non-commutative Khintchine inequalities for $0 < p < 1$
Status: literature already answered, not a new automatic-discovery proof

Source and later answer

The source paper is:

arXiv:0810.5705, Gilles Pisier,

Remarks on the non-commutative Khintchine inequalities for $0 < p < 2$, 2008.

The local copy is included as `source_paper.pdf`.

The later supporting paper is:

arXiv:1412.0222, Gilles Pisier and Éric Ricard,

The non-commutative Khintchine inequalities for $0 < p < 1$.

The local copy is included as `supporting_paper_1412.0222.pdf`.

Open question in the source

On page 3 of arXiv:0810.5705, Pisier states the non-commutative Khintchine inequality

$$|||x|||_q \leq \beta_q \left(\int \left\| \sum_k r_k(t) x_k \right\|_q^q dt \right)^{1/q}, \quad (0.1)$$

where, for $1 \leq q \leq 2$,

$$|||x|||_q = \inf_{x_k = a_k + b_k} \left\{ \left\| \left(\sum a_k^* a_k \right)^{1/2} \right\|_q + \left\| \left(\sum b_k b_k^* \right)^{1/2} \right\|_q \right\}.$$

The paper proves an extrapolation result down to $p \geq 1$, but then says that the argument was not available for $0 < p < 1$, because Step 3 was missing, and explicitly leaves open the validity of (0.1) for $0 < q < 1$.

The same obstruction is identified later in the source after the proof of Theorem 1.1. Pisier uses Junge–Parcet to justify estimate (1.6) when $p \geq 1$, but says that it is not known whether (1.6) still holds for $0 < p < 1$.

Supporting answer

The abstract of arXiv:1412.0222 states that the paper gives a proof of the Khintchine inequalities in non-commutative L_p -spaces for every $0 < p < 1$. Its introduction explicitly refers to the earlier

paper arXiv:0810.5705 as reference [25], says that the $q < 1$ case remained open there, and states that the goal of the paper is to give an explicit estimate and proof for $q < 1$.

The decisive formal statement is Corollary 2.2 of arXiv:1412.0222:

$$K_q \Rightarrow K_p \quad \text{remains valid for every } 0 < p < 1.$$

The same corollary immediately adds that, in particular, the non-commutative Khintchine inequality holds for every $0 < q < 1$.

Claim. *The main open $0 < q < 1$ case of the non-commutative Khintchine inequality (0.1) in arXiv:0810.5705 is already answered by arXiv:1412.0222.*

Proof. The identification is direct. The source asks for the validity of the non-commutative Khintchine inequality (0.1) below exponent 1, and locates the missing step in the non-commutative Holder-type estimate needed for extrapolation. The supporting paper explicitly cites the source, says that exactly the $q < 1$ case remained open there, and then proves the required extrapolation for $0 < p < 1$. Corollary 2.2 states the resulting validity of the non-commutative Khintchine inequality for all $0 < q < 1$.

Therefore the extracted source question has a separate later literature answer. □

Classification and scope

This packet belongs in `solutions/literature_already_answered/`. The supporting paper knows it is answering the missing $q < 1$ case from the source paper and cites that source. This is a retrieval/status result, not a new proof by the run.

The recorded answer is limited to the main non-commutative Khintchine inequality (0.1) for $0 < q < 1$, in the Rademacher setting and the generalized settings covered by the later theorem. This packet does not claim to settle the source paper's distinct questions about best possible sizes of operator subspaces, $\sigma(q)$ -sets versus $\sigma(q)_{cb}$ -sets, or other side questions.

Verification notes

- The source statement is not a same-paper false positive: arXiv:0810.5705 explicitly leaves (0.1) open for $0 < q < 1$.
- The supporting source is separate and later, and explicitly refers back to the source paper's open $q < 1$ case.
- The decisive answer is arXiv:1412.0222, Corollary 2.2, together with the abstract and introduction explaining that the paper proves the missing $0 < p < 1$ non-commutative Khintchine inequalities.

References

- G. Pisier, arXiv:0810.5705, *Remarks on the non-commutative Khintchine inequalities for $0 < p < 2$.*
- G. Pisier and É. Ricard, arXiv:1412.0222, *The non-commutative Khintchine inequalities for $0 < p < 1$.*